



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Sales & business data cleaning & visualisation

Project Domain

[Excel & AI chatbot & Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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1. Introduction

Data analysis plays an important role in business decision-making. Retail stores generate large amounts of sales data every day. This data contains valuable information about customers, products, sales performance, and market trends.

The Vrinda Store dataset contains information about customer orders, sales amounts, product categories, states, and order status. By analyzing this dataset using Microsoft Excel, meaningful insights can be generated to help businesses understand customer behavior and improve sales performance.

This project focuses on cleaning, analyzing, and visualizing sales data using Excel tools such as formulas, pivot tables, and charts.

2. Problem Statement

The organization is unable to clearly track and measure key performance indicators (KPIs), which are essential for evaluating business success. There is limited visibility into important metrics such as Total Sales Revenue, Monthly Sales Growth, Top-Selling Products, Customer Purchase Frequency, and Average Order Value (AOV).

Additionally, the business struggles to identify region-wise sales performance and monitor inventory turnover, resulting in stock shortages or overstocking. The absence of insights into return/refund rates further impacts profitability and customer satisfaction.

Due to these challenges, the business cannot:

- *Identify high-performing products and customers*
- *Track growth trends over time*
- *Optimize pricing and promotional strategies*
- *Improve operational efficiency*

Therefore, there is a strong need to clean, analyze, and visualize the data to generate actionable insights and continuously monitor KPIs for better strategic decision-making.

- *Without proper analysis, businesses may face several challenges such as:*
- *Inability to identify the best-selling products*
- *Lack of knowledge about which states or regions generate the highest revenue*
- *Difficulty in understanding customer purchasing behavior*

- *Poor decision-making due to lack of reliable insights*

The Vrinda Store dataset contains valuable sales information, but it needs to be properly cleaned and analyzed in order to extract meaningful insights. If this data is not processed correctly, businesses cannot fully understand their performance or identify opportunities for improvement.

Therefore, this project aims to apply data analysis techniques using Microsoft Excel to clean, organize, and analyze the dataset. The results of the analysis will help identify important sales trends, customer patterns, and business performance indicators.

3. Objectives of the Project

- *The main objectives of this project are as follows:*
- *To understand the structure of the Vrinda Store dataset, including the number of rows, columns, and types of data available.*
- *To perform data cleaning and preprocessing, including removing duplicate records, correcting errors, and handling missing values.*
- *To organize the dataset in a structured format that is suitable for analysis.*
- *To perform sales analysis using Excel tools, such as formulas, sorting, filtering, and pivot tables.*
- *To identify key business insights, including top-performing products, best-performing states, and monthly sales trends.*
- *To create visualizations such as charts and graphs that clearly present the results of the analysis.*
- *To develop a simple dashboard that summarizes the important insights from the dataset.*
- *To demonstrate the practical application of Excel in business data analysis.*

4. Methodology / Implementation Plan

The project follows a structured approach consisting of several stages. Each stage focuses on a specific step in the data analysis process.

4.1 Data Collection and Exploration

The first step in the project is obtaining the Vrinda Store dataset in Excel format. Once the dataset is collected, the data is explored to understand its structure. This includes identifying the number of rows and columns, understanding the meaning of each field, and examining the types of data stored in the dataset.

During this stage, the dataset is reviewed to detect any inconsistencies or errors that might affect the analysis.

4.2 Data Cleaning and Preprocessing

Raw datasets often contain errors or incomplete information. Therefore, the next step is data cleaning. This process involves removing duplicate records, correcting formatting issues, and handling missing values.

Several Excel tools are used during this stage, including:

- *Remove Duplicates feature*
- *Sorting and filtering options*
- *Data validation*
- *Text formatting tools*
- *Cleaning the dataset ensures that the data is accurate and reliable for analysis.*

4.3 Data Analysis

After cleaning the data, the dataset is analyzed using Excel analytical tools. Various calculations and summaries are performed to identify patterns and trends.

Key Excel functions and tools used in this stage include:

- *SUMIF and COUNTIF functions for calculating totals and counts*
- *Pivot Tables for summarizing large datasets*
- *Sorting and filtering to analyze specific categories or regions*
- *Through these methods, important business insights can be discovered from the dataset.*

4.4 Data Visualization

Once the analysis is completed, charts and graphs are created to visually represent the insights. Visualization makes it easier to understand complex data and communicate findings effectively.

The following types of charts may be used:

- *Bar charts to compare sales by category*
- *Pie charts to show order distribution*
- *Line charts to display monthly sales trends*
- *Column charts to compare sales across states*

These visualizations help present the analysis results in a clear and professional manner.

4.5 Documentation and Reporting

The final stage of the project involves documenting the entire process, including dataset description, data cleaning steps, analysis methods, visualizations, and conclusions. The completed project report will summarize the insights obtained from the dataset and explain how Excel tools were used to perform the analysis.

5. Resources and Requirements

Hardware Requirements & Software Requirements

- *Laptop or personal computer*
- *Minimum 4 GB RAM*
- *Operating System: Windows 10 / 11 or macOS*
- *Microsoft Excel (Microsoft 365 or Excel 2019 or later)*
- *Internet browser for research and references*
- *AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects*

Dataset

- *Dataset Name: Vrinda Store Sales Dataset*
- *Format: Excel (.xlsx)*
- *Data Type: Retail sales transaction data*

6. Expected Outcomes & Conclusion

By completing this project, several outcomes are expected:

- *A cleaned and organized dataset ready for analysis*
- *Identification of important sales trends and patterns*
- *Insights into customer purchasing behavior*
- *Visualization of key business performance indicators*
- *A dashboard summarizing important findings*

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Due to these challenges, the business cannot:

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Therefore, there is a strong need to clean, analyze, and visualize the data to generate actionable insights and continuously monitor KPIs for better strategic decision-making.

The project demonstrates how Excel can be used effectively for data cleaning, analysis, and visualization. By analyzing the Vrinda Store dataset, useful business insights can be generated that help organizations understand their sales performance and improve decision-making.

7. References

- *Microsoft Excel Official Documentation*
- *Kaggle Open Data Repository*
- *Online tutorials on Excel Data Analysis*
- *Educational resources on business data analytics*
- *Edu net Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>*

